Referee report on "Simple local trace formulas for unramified p-adic groups" by David Joyner (version dated 10-97)

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Scope. This report covers the 35-page 1997 revision of the paper (the version whose title-page footnote says it corrects the published version, Pacific J. Math. **154** (1992), 103–129). Page, section and equation numbers refer to that PDF. I checked every displayed formula against the standard sources the paper cites (Macdonald, Casselman, Cartier, Bernstein–Zelevinsky, Arthur, Waldspurger) as far as I can reconstruct them. Where I could not settle a point, I say so and phrase it as a question.

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1 Summary and recommendation

What the paper does. Let G be unramified over a p-adic field F and K hyperspecial. For a spherical f and a test function ϕ supported on the regular elliptic set, the paper wants a "Macdonald–Plancherel" expansion

$$T_\phi(f) = \int_{G(F)\times G(F)} f(h)\phi(g^{-1}hg)\,dg\,dh = \int f^\vee(\chi)\,I(\chi,\phi)\,d\mu(\chi).$$

Equivalently, it wants a spectral formula for elliptic orbital integrals of spherical functions. The method follows Waldspurger's $GL(N)$ paper:

- truncate the integral over $G(F)$ to $G(T) = \bigcup_{a\in\nu_A^{-1}(\mathcal{A}_A(T))} KaK$;

- expand $\Gamma_\chi(g^{-1}hg)$ in matrix coefficients;
- evaluate the truncated inner products of matrix coefficients (§3) using Casselman's asymptotics, Macdonald's formula and an Arthur–Langlands combinatorial lemma;
- pass to a T-independent distribution $I(\chi, \phi)$ (§4).

Section 5 sketches a κ-twisted version for $SL(n)$ and its relation to endoscopic transfer.

Assessment. The strategy is sound and natural, and the main identity (Theorem 4.5) is very plausibly true. As written, though, the paper does not prove it.

- Several standard inputs are misquoted: the constant and the domain of Macdonald's formula, the geometric lemma, Casselman's pairing, and a $\delta_P^{1/2}$ factor.
- The key combinatorial Lemma 3.5 is false as stated.
- Two steps carry the whole argument but are only asserted:
  - that J^T has the form claimed in Proposition 3.6 (the induction only controls $J^U - J^T$);
  - that the limit in Proposition 4.2 is independent of T.

None of this looks fatal: each problem has a standard repair, which I indicate below. But the repairs are substantive, not cosmetic.

Readability. The paper is hard to read. Notation is overloaded:

- T is a maximal torus, a truncation parameter and an elliptic Cartan;
- χ is a character and also $\chi(M)$, a characteristic function;
- Q is a parabolic subgroup and also Macdonald's constant;
- $\hat{\alpha}$, $\alpha^\wedge$, ϖ_α, $\Delta^\vee$ and $R^\wedge$ are used for overlapping objects.

Several conventions are never fixed: which chamber is "negative," normalized versus unnormalized Jacquet modules, and the side on which W_M acts. Lemmas 2.6–2.9 and 4.1 are said to be "not needed," yet in my view they are exactly what is needed to close the gap in §4 (see M8).

Recommendation. Major revision. Specifically:

1. Fix the standard inputs (M1–M4).
2. Replace Lemma 3.5 with a correct vertex (Brion-type) decomposition (M6).
3. Either prove the structural statement in Proposition 3.6 or weaken it to what the induction gives (M7).
4. Make the T-independence in §4 an honest argument, using the $(G, A)^\sim$-family machinery that is already in §2.5 (M8).
5. Update §5 in light of the now-proved fundamental lemma (M9).

2 Major issues

M1. Macdonald's formula (Lemma 2.1, p. 8): constant and domain

(a) **Wrong constant.** With $c_\alpha(\chi) = (1-q^{-1}\chi(a_\alpha))/(1-\chi(a_\alpha))$ in the split case and $\mathrm{vol}(K) = 1$, the constant must be

$$Q = \sum_{w\in W} (IwI : I)^{-1} = \sum_{w\in W} q^{-\ell(w)},$$

not $\sum_w (IwI : I)$.

- *Check at $a = 1$.* For $PGL(2)$, with $z = \chi(a_\alpha)$, $\sum_w c(w\chi) = c(z) + c(z^{-1}) = 1 + q^{-1}$. So $\Gamma_\chi(1) = 1$ forces $Q = 1 + q^{-1}$.
- *Check against Plancherel.* I verified numerically that $\frac{Q}{|W|}\int |c(\chi)|^{-2}\, d\chi = 1 = \|1_K\|^2$ holds for $Q = 1 + q^{-1}$ and fails for $Q = 1 + q$ (it would give q instead of 1).

The same Q appears in Lemma 2.2, Lemma 3.3 and (3.7), so the correction propagates there.

(b) **Wrong domain.** Macdonald's formula holds only for a in the closed dominant (in the paper's convention, "negative") chamber, i.e. $a \in A^-$. It is not valid for all $a \in A(F)$; Γ_χ is W-invariant, while the right side is not. Lemma 2.1 should read "$a \in A^-$." This matters, because the paper later applies the formula on A^- and on Levi subgroups.

(c) **Exponent in c_α (p. 7).** The first factor of $c_\alpha(\chi)$ is printed as $1 - q_{\alpha/2}^{1/2} q_\alpha^{-1}\chi(a_\alpha)$. In Casselman's formula [Cas1, Thm. 3.1] the exponent is $-1/2$. The two agree when G is split ($q_{\alpha/2} = 1$) but differ for non-reduced relative root systems. Please check.

M2. The geometric lemma (Lemma 2.4, Corollary 2.5, Remark 1, pp. 10–11)

(a) **Cosets.** The subquotients of $r_{M,G} \circ i_{G,A}(\chi)$ are indexed by $W_M\backslash W$ (equivalently, by W^M, the minimal-length representatives w with $w^{-1}(\Delta^M) > 0$), not by W/W_M. With left cosets wW_M the character $w\chi$ is not well defined even up to W_M-conjugacy. This is visible later: the proof of Lemma 3.3 (p. 17) writes vw with $v \in W_M$ on the left, which only makes sense for right cosets $W_M w$.

(b) **Irreducibility is false.** The lemma says "each $V_{j+1}/V_j \cong i_{M,A}(w_j\chi)$ is an irreducible M-module." This is false in general: take $M = G$, where the claim would say every unramified principal series is irreducible. Delete "irreducible."

(c) **Remark 1 is false for $M = G$.** For example, $\chi = \delta_B^{1/2}$ on $SL(2)$ is regular, but $i_{G,A}(\chi)$ is a non-split extension of the trivial and Steinberg representations. What is true for regular χ is this: the $W_M\backslash W$-pieces have distinct cuspidal supports $W_M w\chi$, so the Bernstein decomposition of $\mathrm{Alg}\, M$ makes $r_{M,G} \circ i_{G,A}(\chi)$ a *direct sum* of the (possibly reducible) $i_{M,A}(w\chi)$. That is precisely what Lemma 3.1 and Lemma 3.2 need.

(d) **Placement.** Remark 1 sits between Corollary 2.5 and its proof; please move it.

M3. Lemma 3.2: the uniqueness argument (pp. 14–15)

The proof as written has four problems.

(i) "Any bilinear pairing ... is of the form $\sum a_{w,w'}\langle z_w, z'_{w'}\rangle_0$" is false for arbitrary bilinear forms. It needs M-invariance of $\langle\cdot,\cdot\rangle_N$ (Casselman) together with a multiplicity-one statement.

(ii) The displayed actions $z \mapsto \delta^{1/2}(a)(w\chi)(a)z$ are scalars only for $a \in A_M(F)$ (the split center of M), not for $a \in A(F)$.

(iii) Invariance means $\langle \pi(a)z, \tilde{\pi}(a)z' \rangle = \langle z, z' \rangle$. The inequality $\langle r(a)z, z' \rangle \neq \langle z, r(a)z' \rangle$ is not what invariance forbids.

(iv) Casselman's canonical pairing is between V_N and $\tilde{V}_{\bar{N}}$ (opposite parabolic), not $\tilde{V}_N$.

Suggested proof. The pairing lies in $\mathrm{Hom}_M\big(\bigoplus_w i_{M,A}(w\chi) \otimes \bigoplus_{w'} i_{M,A}(w'\chi^{-1}), \mathbb{C}\big)$.

- A non-zero invariant pairing between $i_{M,A}(w\chi)$ and $i_{M,A}(w'\chi^{-1})$ forces a common cuspidal support, so $W_M w\chi = W_M w'\chi$, and hence $W_M w = W_M w'$ by regularity.
- For $w = w'$, Bruhat theory gives $\dim \mathrm{Hom}_M\big(i_{M,A}(w\chi), i_{M,A}(w\chi^{-1})^\sim\big) \leq \#\{v \in W_M : vw\chi = w\chi\} = 1$.

Also, the subscript $\langle f_w, f'_w \rangle_A$ in the statement should be $\langle \cdot, \cdot \rangle_M$. In Remark 3, "$u, u' \in \mathcal{H}(M, M \cap K)$" should read "$u, u'$ are $(M \cap K)$-fixed vectors in $i_{M,A}(w\chi^{\pm 1})$."

M4. Lemma 3.3 and Lemma 3.4: missing $\delta_P^{1/2}$, wrong domain, proof errors (pp. 15–17)

(a) **Missing modulus factor.** With normalized induction and Jacquet functors, Casselman's asymptotic is

$$\langle \pi(a)f, f' \rangle = \delta_P^{1/2}(a) \langle \pi_N(a) f_N, f'_{\bar{N}} \rangle_N.$$

The statement of Lemma 3.3 omits $\delta_P^{1/2}(a)$. For $M = A$ the lemma must reproduce Macdonald's $\delta_B^{1/2}(a)$, and the proof itself inserts $\delta(a)^{1/2}$. Similarly, Lemma 3.4 needs $\delta_P(a)$ and the coefficients $c(w, M, \chi)c(w', M, \chi')$ in its statement. Its proof has them, but the statement and (3.13) do not.

(b) **Domain.** As stated, for $a \in A^-(\epsilon)$ (deep in *every* simple direction), the version for general M adds nothing beyond $M = A$. The inductive proof of Proposition 3.6 actually uses it on the region

$$A_P^-(\epsilon) = \{a \in A^- : \ |\alpha(a)| \leq \epsilon \ \ \forall \alpha \in \Delta \smallsetminus \Delta^M\},$$

which is deep only in the N-directions. This is Casselman's Proposition 4.3.3 [Cas2], and it is the correct hypothesis for Lemmas 3.3–3.4. Please restate them this way.

(c) **Errors in the proof of Lemma 3.3.**

- The display $\Gamma_\chi(a) = \sum_w c(w, A, \chi)\delta(a)(w\chi)(a)(w\chi^{-1})(a)c_w(\chi)c_w(\chi^{-1})$ contains $(w\chi)(a)(w\chi^{-1})(a) = 1$. Only the first argument is translated, so the factor $(w\chi^{-1})(a)$ must go, and $\delta(a)$ should be $\delta(a)^{1/2}$.
- In the general-M computation, Macdonald's formula for G is missing Q_G^{-1}, and the formula for M is missing Q_M^{-1} on the first line where it is used.
- The final display drops the factor $Q_G^{-1}Q_M$ that appears in the statement.
- "for any $v \in W/W_M$" should be $v \in W_M$.

(d) **Undefined notation.** c_w^M is never defined. Since w is a minimal-length representative, $c_w^M = c_w$; please say so. Also, χ' in the lemma should be χ^{-1}.

(e) **Remark 5.** The right side reads $\kappa(g)\int_{KaK}\cdots dg$, with the integration variable left outside. It should be $\kappa(a)$; this holds because κ is unramified, hence constant on KaK.

(f) **Levels.** For Lemma 3.4 one needs $\pi(k)u$ to have the same level as u. Choose the level subgroups normal in K (principal congruence subgroups).

With (a)–(b) fixed, the formula for $c(w, M, \chi)$ in Lemma 3.3 is correct. I checked it against Macdonald's formula for G and M; the ratio $c(vw\chi)/c_M(vw\chi)$ is independent of $v \in W_M$, as required.

M5. The claim $\nu_A^{-1}(\mathcal{A}_A(T)^c) \subset A^-(\epsilon)$ (p. 18) is false

The complement of the truncated region contains points near walls.

- *Counterexample.* In $SL(3)$, $a_n = \nu_A^{-1}(3n\varpi_1^\vee)$ with $3\varpi_1^\vee = 2\alpha_1^\vee + \alpha_2^\vee$ lies outside $\mathcal{A}_A(T)$ for $n \gg 0$, yet $|\alpha_2(a_n)| = 1 > \epsilon$.
- *Consequence.* The consequence $G(T)^c \subset KA^-(\epsilon)K$ fails as well.
- *Repair.* This is harmless once M4(b) is adopted: the $\xi(M, D, T, U)$-decomposition in Proposition 3.6 exists precisely to cut the complement into pieces where the appropriate $A_P^-(\epsilon)$-asymptotic applies. Delete the claim and say this instead.
- *Related step.* On p. 22 the inequality $\langle\beta, X\rangle \geq \langle\beta, T\rangle$ needs justification: $\langle\beta, X^M\rangle \leq 0$, so one only gets $\langle\beta, X\rangle > \langle\beta, T_M\rangle + \langle\beta, X^M\rangle$. The bound is fine up to a constant depending on T^M, which suffices for T sufficiently regular, but this should be written out.

M6. Lemma 3.5 and (3.7) (pp. 18–19)

(a) **Parametrization.** The proof writes $\chi(a) = \prod_\alpha q^{s_\alpha\langle a,\alpha\rangle}$. This treats $a \mapsto (\langle a, \alpha\rangle)_{\alpha\in\Delta}$ as a bijection $X_*(A) \to \mathbb{Z}^\ell$, which holds only when $X_*(A)$ is the coweight lattice (adjoint G). The generators of the dominant cone are then $\varpi_\alpha^\vee(\pi)$, not $a_\alpha = \alpha^\vee(\pi)$. In general the dominant cone in $X_*(A)$ is not unimodular, and the generating function has a numerator from a fundamental parallelepiped. Either restrict to adjoint G or state the general form.

(b) **The "in fact" clause is false, already for $SL(2)$.** With $T = T_\alpha\alpha^\vee$ and $z = \chi(a_\alpha)$, one gets $\nu_A^{-1}(\mathcal{A}_A(T)) = \{\nu_A^{-1}(n\alpha^\vee) : 0 \leq n < T_\alpha\}$. The $\omega = \Delta$ piece is

$$\sum_{n=1}^{T_\alpha - 1} z^n = \frac{z - z^{T_\alpha}}{1 - z},$$

so $F_{\Delta,T}(\chi) = z - z^{T_\alpha}$. That is not of the form $z^{T_\alpha}F_\Delta(z)$. Also, $F_{\emptyset,T} = 1$ carries no $\chi(T)$ factor at all.

(c) **The correct statement** is a vertex (Brion–Lawrence) decomposition:

$$\sum_{a\in\nu_A^{-1}(\mathcal{A}_A(T))} \chi(a) = \sum_{\omega\subset\Delta} \chi(T_\omega)\, R_\omega(\chi),$$

where T_ω is the vertex of the truncated polytope determined by ω and each R_ω is a rational function independent of T. In rank one this is $\frac{1}{1-z} - \frac{z^T}{1-z}$. This is the form that the later $T \to \infty$ argument needs: the $\omega = \emptyset$ term is T-independent, and the others oscillate in T. As written, the face decomposition (3.5) in the proof does not deliver it, because the truncation $\langle\varpi_\alpha, a - T\rangle < 0$ is a simplex-type condition, not a box, in the coordinates n_α.

(d) **(3.7).** Macdonald gives $\mathrm{meas}(KaK) = \delta(a)^{-1}W(q^{-1})/W_a(q^{-1})$, where W_a is the stabilizer of a. So Q_a is rational, not an integer; for $SL(2)$, $a \neq 1$ gives $Q_a = 1 + q^{-1}$.

(e) **Remark 6.** The poles statement inherits (a).

M7. Proposition 3.6: what the induction actually proves

(a) **A T-independent constant is left over.** The inductive step computes the pieces $M \neq G$ of $J_D^U = \sum_M J_D^{T,U}(M,\cdot)$, where the $M = G$ piece is J_D^T itself, and then applies the combinatorial lemma (3.23). This shows

$$J^T - F(T) = J^U - F(U) \qquad (U \geq T \geq T_0),$$

with $F(T) = c_G(-1)^{\mathrm{rk}G} \sum_{a \in \mathcal{A}_A(T)} \gamma_a$. Hence $J^T = F(T) + C$, with C independent of T. The argument never shows $C = 0$, and C is exactly the contribution of the compact core, where no asymptotics are available. Please either:

- supply the missing step (e.g. via the exponential-polynomial structure in T from M6(c), identifying the T-independent term); or
- state the proposition as an identity for $J^T - J^U$.

(b) **Is the structural claim plausible for arbitrary u, u'?** $J^T(u,u')$ is a bilinear form on $V_B(\chi^{-1}) \times V_B(\chi'^{-1})$ that is invariant only under the diagonal K-action. The space of such forms is infinite-dimensional. A priori there is no reason it should lie in the span of the $|W|^2$ forms $\langle T_w u, T_{w'} u' \rangle$ with coefficients independent of u, u'. For K-fixed (or Iwahori-fixed) u, u' the exact Macdonald/Casselman formulas hold on all of A^-, and the claim is credible. For general u, u' I would want to see a proof. (This only matters in §4, where u_i runs over a basis.)

(c) **Base case.** For a torus, definition (3.3) makes $\mathcal{A}_A(T)$ a single point (everything is taken modulo the center), not the box $0 < \nu_F(a_i) < T_i$. The box belongs to the M-relative truncation. Please align the definitions. The sign $(-1)^{\mathrm{rk}(M)}$ in (3.20) needs a definition of rk ($\dim \mathfrak{a}_M/\mathfrak{a}_G$?) and a check in the base case.

(d) **Reindexing on p. 24.** I would have expected $\sigma = vw$ with $\ell(vw) = \ell(v) + \ell(w)$, so that $T_v^M R_M T_w = R_M T_{vw}$, rather than $\sigma = w_\ell^G w_\ell^M vw$. The latter would come from Waldspurger's opposite-parabolic conventions. Whichever is intended, it must be used consistently with the characters $\sigma\chi$ appearing in γ_a.

(e) **Notation clash.** $\chi(M)$ for a characteristic function, next to the character χ, is confusing; please rename it (e.g. $\mathbf{1}_M$).

M8. Proposition 4.2 and Theorem 4.5: T-independence is not established

(a) **What is shown and what is not.** For $T \geq T(f,\phi)$, Fubini over the compact set $G(T)$ correctly gives

$$I(f,\phi) = \int f^\vee(\chi)\, I^T(\chi,\phi)\, d\mu(\chi).$$

But $I^T(\chi,\phi)$ itself depends on T: enlarging T adds the integral of $\Gamma_\chi(g^{-1}hg)\phi(h)$ over a shell, which is generally non-zero and oscillates in χ. Indeed $g \mapsto \int \Gamma_\chi(g^{-1}hg)\phi(h)\, dh$ is not compactly supported, which is why a truncation was introduced in the first place. The

sentence "the claim that the result is independent of T follows from Proposition 3.6" is therefore unsupported; Proposition 3.6 says nothing of the kind.

(b) **Suggested argument.** Write $I^T(\chi,\phi) = I(\chi,\phi) + \sum(\text{terms of the form } (w\chi/w'\chi)(T)\cdot S(\chi))$, where $I(\chi,\phi)$ is the T-independent ($\omega = \emptyset$) part. Show that the coefficients S are *smooth* on the unitary torus after summing over w, w'. This is exactly the $(G,A)^{\sim}$-family statement of Lemmas 2.7, 2.9 and 4.1, which removes the apparent poles of $\tilde{\theta}(w\chi/w'\chi)^{-1}$. Then $\int f^{\vee}(I^T - I)\,d\mu$ tends to 0 as $T \to \infty$ (Riemann–Lebesgue on the compact torus). Since the left side is constant for $T \geq T(f,\phi)$, it vanishes. So, contrary to the remarks on pp. 13 and 26, those lemmas *are* needed. This is also how Waldspurger and Arthur argue.

(c) **Definition of $I(\chi,\phi)$.** Proposition 4.2 never defines $I(\chi,\phi)$ independently of the displayed limit. Also, Q^T in (4.1) already contains the factor $\tilde{\theta}(w\chi/w'\chi')$, which the proposition then divides out again. Please define $I(\chi,\phi)$ explicitly (e.g. as the T-independent term) and state Proposition 4.2 as "for all $T \geq T(\phi)$, $I^T(\chi,\phi) = I(\chi,\phi) + (\text{oscillatory terms})$."

(d) **The trace identity (p. 29).** The second argument in (4.7) should be u_j, not u_j^*: $J^T(\chi,\chi^{-1},u_i^*,u_j)$. Moreover $T_w^* T_{w'}$ only makes sense as an operator on one space after identifying all $V_B(\cdot)$ with $C^\infty(K \cap B\backslash K)$ (the compact picture); please say so. The "$(A(F)\cap K)$-bi-invariant orthonormal basis" should be a basis adapted to a filtration by normal open subgroups $K_n \subset K$. That is what makes the sums over i,j finite, since $\pi(g)\Phi_{K,\chi}$ is K_n-fixed for g in a compact set.

(e) **Admissibility.** G-admissibility of $\phi \mapsto I(\chi,\phi)$ via [HC, §14] is fine once (b) is in place.

(f) **Geometric side.** In (4.10)–(4.11) the factor $\mathrm{meas}(T(F))$ appears in one line and not the next. "$\Delta(t)^2$" should be defined; it is the Weyl discriminant $|D(t)|$.

M9. Section 5 (κ-orbital integrals on $SL(n)$)

(a) **Sign of κ in (5.1).** The change of variables $h \mapsto ghg^{-1}$ followed by $g \mapsto g^{-1}$ produces $\kappa(g)^{-1}$. So the second line of (5.1) should carry κ^{-1}, and the last line should read $\Phi_\phi^{\kappa^{-1}}$. This is consistent with (5.4), where $\Phi_f^{\kappa^{-1}}$ appears. In (5.3), the integrand defining Φ_ϕ^κ should contain ϕ, not f. Lines 3–4 of (5.1) have a stray "dg."

(b) **Weyl integration on $SL(n)$ versus $PGL(n)$.** (5.2) integrates over $T(F)\backslash G'(F)$, while ψ lives on $G(F) = SL(n,F)$. One must account for the finite group $G'(F)/G(F)T'(F)$, which permutes the $SL(n)$-classes inside a $PGL(n)$-class. This is where the κ-weights come from, and it should be written out.

(c) **Endoscopic groups.** For κ of order $m \mid n$, the elliptic endoscopic group of $SL(n)$ is

$$H = \ker\big(N_{E/F} \circ \det\big) \subset \mathrm{Res}_{E/F} GL(n/m), \qquad [E:F] = m,$$

not $\mathrm{Res}_{E/F} GL(m)$. For $m = n$, H is the norm-one torus (a Cartan of $SL(n)$), not a Cartan of $GL(n)$. In (5.11) the right side should be a *stable* orbital integral $S\Phi_{f_H}$. Also, H is reductive, not semisimple, so Theorem 4.5 must be stated with $G(F)^1$ for reductive groups.

(d) **The fundamental lemma is now a theorem.** "It has been conjectured" is out of date:

- for the unit element for $SL(n)$: Kazhdan for κ of order n, and Waldspurger [Canad. J. Math. 43 (1991)];

- reduction to unit elements: Hales [Canad. J. Math. 47 (1995)];
- in general: Ngô [Publ. Math. IHÉS 111 (2010)], with Waldspurger's transfer from Lie algebras to groups and Laumon–Ngô for unitary groups.

The final question (does $\phi \mapsto I^\kappa(\iota(\chi_H), \phi)$ transfer to $\phi_H \mapsto I_H(\chi_H, \phi_H)$?) is formally a consequence of the FL for the Hecke algebra together with Theorem 4.5 for G and H. It would make a nice concluding proposition rather than a question.

(e) **Parenthetical on p. 32.** The parenthetical "if $\phi \in \mathcal{H}(G, K)$ then $\phi^T = i_*(\phi)(t)$" conflicts with the standing assumption that ϕ is supported on G_{ell}. It should be phrased for a general ϕ, or refer to f.

3 Minor mathematical corrections (by location)

Abstract / p. 2

The abstract says "semi-simple," while §2.1 and §3 allow reductive groups (and the induction needs Levi subgroups, which are reductive). State the main theorem for reductive G with $G(F)^1$. "A distribution on the support of the Plancherel measure (regarded as a compact complex analytic variety)": the support is the compact real torus of unitary unramified characters modulo W. Its complexification is the complex torus on which $I(\chi, \gamma)$ is rational. Please separate the two.

p. 2

"compute, in §§2–3": the computation is in §§3–4.

p. 3

"In §4, we examine ... the κ-stable version": that is §5. "(see Theorem 4.5 below)" is fine.

p. 4, §2.1

T should be the centralizer of A (a maximal torus defined over F, containing A, inside B). It is not an arbitrary maximal torus.

p. 4

"$\Delta^\vee$ the set of dual roots $\{\varpi_\alpha\}$" conflicts with p. 5, where $(\Delta_P^Q)^\vee$ is a basis of $\mathfrak{a}_P^Q$ (coroots) and $(\Delta_P^Q)^\wedge$ a basis of $(\mathfrak{a}_P^Q)^*$ (fundamental weights). Follow Arthur: $\Delta^\vee = \{\alpha^\vee\}$ and $\hat{\Delta} = \{\varpi_\alpha\}$.

p. 5

"$X^*(A)/R$ trivial iff adjoint, $X_*(A)/L$ trivial iff simply connected" is correct for the absolute root datum, or for split G. For relative roots of non-split groups it can fail; e.g. quasi-split $SU(3)$ and $PU(3)$ both have $X^*(A) = R$.

p. 5

"$P_\emptyset = B$, $M_\emptyset = A$": $M_\emptyset = Z_G(A) = T$, which equals A only for split G.

p. 5

$\hat{\tau}_P^Q$ is defined by exactly the same condition as τ_P^Q. It should be $\{X : \varpi(X) > 0,\ \varpi \in \hat{\Delta}_P^Q\}$.

p. 6

Property (b), that the a_α generate $A(F)/A(F) \cap K$ freely, holds only for simply connected G. In general they generate a sublattice of finite index $[X_*(A) : L]$.

p. 6
"$\langle \alpha^\vee, \beta \rangle = 2\delta_{\alpha\beta}$" is false; the left side is the Cartan matrix. The dual basis to $\Delta^\vee$ is the set of fundamental weights, with $\langle \alpha^\vee, \varpi_\beta \rangle = \delta_{\alpha\beta}$.

p. 7
(2.6)–(2.7):

- $X^*(A)_{\mathbb{C}}/X_*(A)^\perp$ is the group of all complex characters of $X_*(A)$ (a complex torus); the Pontryagin dual is its compact real form.
- The lattice in (2.7) should be $\frac{2\pi i}{\log q} X^*(A)$, the dual of $X_*(A)$, scaled by $2\pi i/\log q$. It is not $\{\lambda : \lambda(\hat{\alpha}) \in \mathbb{Z},\ \alpha \in \Delta\}$, which is the right lattice only in the adjoint case and has the wrong scaling.
- "$\mathcal{A}^*_{A,\mathbb{R}}$ the complex dual" is self-contradictory notation.
- Please define $\mathbb{C}^{\times 1}$ (the unit circle).

p. 9
Lemma 2.2: the measure lives on $i\mathfrak{a}^*/L$ (the unitary characters). The second statement of Corollary 2.3 has $f_1^\vee \overline{f_1^\vee}$; it should be $f_1^\vee \overline{f_2^\vee}$. Polarization needs both $f_1 + f_2$ and $f_1 + if_2$.

pp. 11–13
Three different definitions of χ_λ appear: $\langle \nu_A^{-1}(a), \lambda \rangle$, then $\langle \lambda, \nu_A(a) \rangle$. The first is not even well typed, and neither is a character. It should be $\chi_\lambda(a) = q^{-\langle \nu_A(a), \lambda \rangle}$ (or $e^{\langle \nu_A(a), \lambda \rangle \log q}$, with the sign fixed once). In the hyperplane condition, $\langle \alpha, \lambda \rangle$ should be $\langle \alpha^\vee, \lambda \rangle$.

p. 12
Lemma 2.7: "$c_P(\chi)$" should be $\tilde{c}_P(\chi)$, and "$\tilde{\theta}_B$ is defined in (2.13)" should cite (2.17). Lemma 2.8: $R^\wedge$ and $\hat{R}$ name the same lattice twice. For the shift-invariance to make sense it must be (a scaled copy of) $X^*(A)$, i.e. L from (2.7).

p. 13
Lemma 2.9: the right side $\sum_r c_P^r(\cdots)$ has a dangling P. Presumably $\{c_P^r\}_P$ is a (G, A)-family for each r, and the sum is of c_A^r.

p. 13
(3.1): "ϵ independent of χ" is right, but ϵ depends on the level of f, f' (as Lemma 3.3 later says); state it here.

p. 14
The proof of Lemma 3.1: the displayed lines should read $i_{M,A}(w\chi)(a) R_M T_w f$ (not $i_{M,A}(\chi)_N(a) R_M T_w v$).

p. 18
(3.3) uses $\langle \hat{\alpha}, a - T \rangle$, while the proof on p. 19 uses $\langle \alpha^\vee, a - T \rangle$, which pairs X_* with X_*. Use ϖ_α consistently.

p. 26
Lemma 4.1: "$\mathrm{tr}[R_{w_P} i_{G,A}(\chi_\lambda)]$" is missing the argument (ϕ). "Let T, χ, χ' be as in section 2" should say section 3.

p. 27
"In §4, we will also consider" should say §5. In Proposition 4.3, $\kappa\chi$ mixes a character of $G'(F)$ with one of $A(F)$; say "$\kappa|_{A(F)}\chi$," and note that χ is a character of $A(F)$, not $A'(F)$.

p. 28
Remark 9 refers to "part (a)/(b)," but Propositions 4.2 and 4.3 have no parts. It should say "we prove 4.2; 4.3 is similar."

p. 30
"the Plancherel formula (see Lemma 2.4)" should cite Corollary 2.3.

4 Typos and wording

interprete → interpret (p. 3); Plancheral → Plancherel (pp. 4, 18, 20, 25); fiexed → fixed (p. 7); decomosition → decomposition (p. 5); isomophism → isomorphism (p. 7); "Lemma 2.1. f χ is" → "If χ is" (p. 8); choosen → chosen (passim); "on of the two roots" → "one of" (p. 11); occuring → occurring (p. 13); "A truncated inner products" → "A truncated inner product" (§3.2 title); "in the limit aa" → "as" (p. 26); "it's dual" → "its dual" (p. 21); "not worst than" → "no worse than" (p. 24); $u'' \to u'$ (pp. 17, 23); stray "dk" at the end of (3.19); "Lemma 4.4, ." (p. 31); "Let W_M denotes" → "denote" (p. 10); "one wants to find a both" → "to find both" (p. 3); "relatively explicit for $I(\chi, \gamma)$" is missing a noun (p. 4); "NARC grant" (please spell out the agency); "Ad-finite in the sense of [K]" (Kazhdan calls these compactly generated supports; please give the precise reference).

5 Readability recommendations

1. **Fix conventions once (new §2.0).** State:
   - normalized induction and Jacquet functors;
   - which chamber A^- is;
   - $W_M \backslash W$ and minimal-length representatives;
   - the sign convention for χ_λ;
   - the measure normalizations ($\mathrm{vol}(K) = 1$, $\mathrm{vol}(A \cap K) = 1$).
2. **Rename overloaded symbols.** Use $\mathbf{T}$ (or Y) for the truncation parameter, T for tori, $\mathbf{1}_M$ for characteristic functions, Q_G for Macdonald's constant, and $\hat{\Delta}$ for fundamental weights.
3. **Add a road map.** The proof of Theorem 4.5 passes through Propositions 3.6 → 4.2 → 4.5; a half-page overview in the introduction would help a lot.
4. **Work out $SL(2)$ / $PGL(2)$ in full.** An appendix computing $I(\chi, \gamma)$ explicitly for elliptic γ would check the normalizations end to end and would show readers what the "rational function" $I(\chi, \gamma)$ looks like. I would also recommend checking it numerically against a direct orbital-integral computation.
5. **Move §2.5 next to where it is used.** Once M8 is fixed, the $(G, A)^\sim$-family material is used, so it belongs in §4.
6. **Footnotes on the publication history.** The footnotes saying which parts "did not appear in the published version" are useful historically. In a revised version they could become a single paragraph in the introduction.

6 Bibliography corrections

[A]
M. Assem, Compositio Math. **78** (1991), **37–78** (not 317–378).

[Art1]
Published: J. Arthur, "Towards a local trace formula," in *Algebraic Analysis, Geometry and Number Theory*, Johns Hopkins Univ. Press, 1989, 1–24. Also cite the definitive paper: "A local trace formula," Publ. Math. IHÉS **73** (1991), 5–96.

[BD]
The "—" attributes this to Assem. The author is J. Bernstein (rédigé par P. Deligne), "Le centre de Bernstein," in *Représentations des groupes réductifs sur un corps local*, Travaux en Cours, Hermann, 1984. (It is also never cited in the text.)

[Car]
Add: Proc. Sympos. Pure Math. **33**, Part 1 (1979), 111–155.

[Cas2]
Casselman's notes remained unpublished; cite the date (1974/1995 draft).

[HC]
Harish-Chandra, *Admissible invariant distributions on reductive p-adic groups*, Queen's Papers in Pure and Appl. Math. **48** (1978), 281–347. A revised edition, with notes by S. DeBacker and P. Sally, appeared as AMS University Lecture Series 16 (1999).

[K]
D. Kazhdan, "Cuspidal geometry of p-adic groups," J. Analyse Math. **47** (1986), 1–36.

[Kaz]
LNM **1041** (1984), 209–249.

[M]
Macdonald's notes are from **1971** (Publ. Ramanujan Inst. 2), not 1981. A new edition, with Anne-Marie Aubert, has just appeared in Springer LNM **2392** (2026).

[W]
J.-L. Waldspurger, "Intégrales orbitales sphériques pour $GL(N)$ sur un corps p-adique," Astérisque **171–172** (1989), 279–337.

[Wa]
Proc. ICM Zürich 1994, Birkhäuser 1995.

[Wald]
Correct title: "Sur les germes de Shalika pour les groupes linéaires," Math. Ann. **284** (1989), 199–221. (It is not cited in the text; the reference to the germ expansion on p. 4 cites [W].)

[J]
"In preparation." If this never appeared, remove the reference or describe the result inline.

A Plan for the revised version

For the transcribed revision I propose:

- keep your text and numbering as close as possible to the original;
- make the corrections above silently where they are clear-cut (M1–M4, M9(a), §3);
- mark each substantive change with a dated "Remark (2026)";
- for the two genuine gaps (M7(a) and M8), either insert the repair sketched above as a new lemma, or, where I cannot fully verify the repair, state precisely what is proved and flag the rest.

Glossary (proposed entries). unramified group; hyperspecial maximal compact subgroup; Iwahori subgroup; relative root system and reduced roots; coroots and fundamental (co)weights; $\mathfrak{a}_P$, $\mathfrak{a}_P^Q$, H_M, $G(F)^1$; τ_P^Q, $\hat{\tau}_P^Q$; unramified character and the complex/compact tori of characters; unramified principal series $i_{G,A}(\chi)$; normalized Jacquet functor $r_{M,G}$; geometric lemma; regular character; intertwining operator T_w and c-functions c_α, c_w, $c(\chi)$; zonal spherical function Γ_χ; Satake transform $f^\vee$; Macdonald's formula; spherical Plancherel measure; Casselman's canonical pairing; regular elliptic element; orbital integral; κ-orbital integral; stable orbital integral; Weyl discriminant; (G,M)-family and $(G,A)^\sim$-family; truncation parameter; Langlands' combinatorial lemma (Γ_M^G); G-admissible distribution; endoscopic group; transfer factor; fundamental lemma.

Literature review: outline with verified references. *Before 1997.*

- *Spherical harmonic analysis:* Satake (1963); Macdonald (1971); Casselman (1980); Bernstein–Zelevinsky (1977); Harish-Chandra's Plancherel theorem and admissible invariant distributions (1970s).
- *Arthur's global-to-local program:* the trace formula, Duke 1978, and its invariant form, Ann. Math. 1981; "Towards a local trace formula" (1989); and the local trace formula itself, Publ. Math. IHÉS **73** (1991), 5–96.
- *Elliptic harmonic analysis:* Kazhdan's cuspidal geometry and elliptic pairing (J. Analyse Math. **47**, 1986); Bernstein–Deligne–Kazhdan (1986); Arthur, "On elliptic tempered characters," Acta Math. **171** (1993), 73–138.
- *Spherical orbital integrals:* Waldspurger's computation for $GL(N)$ (Astérisque 171–172, 1989); his Shalika germs paper (Math. Ann. 1989); Assem's unipotent orbital integrals (Compositio 1991).
- *Endoscopy:* Kottwitz (rational conjugacy, 1982; elliptic singular terms, 1986); Langlands–Shelstad transfer factors (Math. Ann. **278**, 1987); Kazhdan, "On lifting" (LNM 1041, 1984); Waldspurger's fundamental lemma for $SL(n)$ (Canad. J. Math. **43**, 1991, 852–896); Clozel and Labesse on stable base change (Duke **61**, 1990); Hales' reduction to unit elements (Canad. J. Math. **47**, 1995, 974–994).
- *Lie algebras:* Waldspurger's local trace formula for p-adic Lie algebras (J. reine angew. Math. **465**, 1995, 41–100).

After 1997.

- *Transfer for $SL(n)$:* Courtès (Mém. SMF **69**, 1997).
- *Transfer in general:* Waldspurger, "Le lemme fondamental implique le transfert" (Compositio **105**, 1997, 153–236).

- *Elliptic theory via the building:* Schneider–Stuhler on Euler–Poincaré functions and elliptic characters (Publ. Math. IHÉS **85**, 1997, 97–191); Reeder's elliptic pairings for Iwahori-spherical representations (Compositio, 2001). These are directly relevant to the elliptic part of the unramified principal series studied here.
- *Plancherel formula:* Waldspurger's exposition of Harish-Chandra's Plancherel formula (J. Inst. Math. Jussieu **2**, 2003, 235–333).
- *The fundamental lemma:* Laumon–Ngô (Ann. Math. **168**, 2008, 477–573); Ngô's proof (Publ. Math. IHÉS **111**, 2010, 1–169).
- *Twisted versions:* Lemaire–Mœglin–Waldspurger's reduction to unit elements for twisted endoscopy (Ann. Sci. ENS **51**, 2018); Mœglin–Waldspurger, *La formule des traces locale tordue* (Mem. AMS **251**, no. 1198, 2018).
- *Local relative trace formulas:* Feigon (Pacific J. Math. **260**, 2012); Delorme–Harinck (Pacific J. Math. **291**, 2017); Delorme–Harinck–Souaifi; Beuzart-Plessis's local trace formula for Gan–Gross–Prasad (Astérisque **417**, 2020).
- *New edition of Macdonald:* with A.-M. Aubert (LNM **2392**, 2026).